# AYOUB AKOUHAR

Software Engineering Student | Systems & Backend Development

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## PROFESSIONAL SUMMARY

Software Engineering student at 1337 School (42 Network) with hands-on experience in C/C++, UNIX/Linux, TCP/IP networking, and backend development. Built HTTP and IRC servers, UNIX system tools, and containerized services with Docker Compose. Strong foundation in data structures, algorithms, debugging, and collaborative code evaluation. Currently learning Python, FastAPI, and application security. **Available for a 6-month internship.**

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Bash, TypeScript, JavaScript

**Systems & Networking:** UNIX/Linux, TCP/IP, HTTP, Sockets, Non-blocking I/O, poll/epoll, Process Management

**Backend & Databases:** FastAPI (Learning), REST API Fundamentals, HTTP Parsing, Routing, SQLAlchemy, Pydantic, PostgreSQL

**Containers & Tools:** Docker, Docker Compose, Nginx, Git, GitHub, GDB, Valgrind, Makefile, Linux CLI

**Computer Science:** Data Structures & Algorithms, OOP, Memory Management, Event-Driven Architecture, Debugging

**Security Fundamentals:** HTTP Protocol Analysis, Web Application Security and Penetration Testing (Learning)

## TECHNICAL PROJECTS

### HTTP Web Server | C++, HTTP, TCP/IP

- **Engineered** an HTTP web server in C++ supporting request processing, HTTP responses, and static file serving.
- **Implemented** HTTP parsing, URL routing, header processing, status codes, and concurrent connections using I/O multiplexing.

### FT_IRC - Concurrent TCP Server | C++, Socket Programming

- **Developed** an event-driven, multi-client IRC server with authentication, channels, and real-time message routing.
- **Implemented** non-blocking socket I/O and multiplexing for concurrent connections and managed client lifecycles.

### Inception - Containerized Web Infrastructure | Docker, Docker Compose, Nginx, MariaDB

- **Containerized** Nginx, WordPress (PHP-FPM), and MariaDB services with Docker and Docker Compose.
- **Configured** HTTPS with TLS, container networking, persistent volumes, and service startup dependencies.

### Minishell - UNIX Command Interpreter | C, POSIX

- **Built** a Bash-inspired shell supporting tokenization, command parsing, pipelines, redirections, and variable expansion.
- **Managed** processes, file descriptors, and signals with fork, execve, dup2, pipe, and waitpid; debugged execution errors.

### Backend API Development | Python, FastAPI, SQLAlchemy (In Progress)

- **Developing** a friendship-management backend module using FastAPI, SQLAlchemy, PostgreSQL, and Pydantic.
- **Implementing** request validation and response schemas while learning REST API and database design principles.

### Push_swap - Algorithm Optimization | C, Data Structures

- **Designed** a stack-sorting program focused on minimizing operations through algorithmic problem solving.

### NetPractice - Networking | TCP/IP

- **Analyzed** subnetting, routing tables, and IP addressing to troubleshoot network connectivity.

## EDUCATION

### 1337 School - 42 Network | Software Engineering Program

- **Developed** proficiency in C/C++, UNIX systems, networking, and algorithms through intensive, project-based learning.
- **Participated** in peer code reviews, technical discussions, and collaborative debugging.

## ADDITIONAL LEARNING

- **Studying** FastAPI, modern backend architecture, and introductory web application security concepts.
- **Practicing** data structures and algorithmic problem-solving on LeetCode.